

database.sql

Separate PDF version of **database.sql** for the Daily Habit Monitoring System project.

```
-- Daily Habit Monitoring System
-- MySQL database and table creation

CREATE DATABASE IF NOT EXISTS habit_monitoring;
USE habit_monitoring;

CREATE TABLE IF NOT EXISTS users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50) UNIQUE NOT NULL,
    password VARCHAR(100) NOT NULL
);

CREATE TABLE IF NOT EXISTS habits (
    habit_id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    habit_name VARCHAR(100) NOT NULL,
    category VARCHAR(50),
    target VARCHAR(100),
    FOREIGN KEY (user_id)
        REFERENCES users(user_id)
        ON DELETE CASCADE
);

CREATE TABLE IF NOT EXISTS habit_records (
    record_id INT AUTO_INCREMENT PRIMARY KEY,
    habit_id INT NOT NULL,
    record_date DATE NOT NULL,
    status VARCHAR(20) NOT NULL,
    FOREIGN KEY (habit_id)
        REFERENCES habits(habit_id)
        ON DELETE CASCADE
);

-- Example INSERT
-- INSERT INTO users(username, password)
-- VALUES ('student1', '1234');

-- Example SELECT
-- SELECT * FROM habits;

-- Example JOIN
-- SELECT hr.record_id, h.habit_name,
--        hr.record_date, hr.status
-- FROM habit_records hr
-- JOIN habits h ON hr.habit_id = h.habit_id;

-- Example GROUP BY
-- SELECT h.category, COUNT(*) AS total_records
-- FROM habit_records hr
-- JOIN habits h ON hr.habit_id = h.habit_id
-- GROUP BY h.category;
```